

Last class: Algebra of matrices

$$2 \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

if M, N are matrices $M+N$ is only defined if they have the same dimension

$$f(x) = e^x \quad g(x) = x^2 + x + 1 \quad f, g: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) + 3g(x) = e^x + x^2 + x + 1$$

Composition $f(g(x)) = (f \circ g)(x) = e^{x^2+x+1}$
+ is commutative for #'s, func's and matrices
composition is not: $(g \circ f)(x) = e^{2x} + e^x + 1$ not the same

if $L: A \rightarrow B$, $K: C \rightarrow D$ are linear, the composition

$(K \circ L)$ is only defined if $B=C$ $A \xrightarrow{L} B=C \xrightarrow{K} D$

the matrix for $(K \circ L)$ is the matrix multiplication of K and L in that order

$$[K][L]$$

Ex: consider $\mathbb{R}^2 \xrightarrow{A} \mathbb{R}^3 \xrightarrow{B} \mathbb{R}^3$

$$A = \begin{bmatrix} 1 & -3 \\ 0 & 2 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$$

$$(B \circ A) = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ 0 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -3+4 \\ 1 & -3+2 \\ 0 & 4+1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 5 \end{bmatrix}$$

Ex: $f(x) = x^2 + x$ $I(x) = x$ $(I \circ f) = f$ $(f \circ I) = f$

The identity function $\mathbb{R}^n \rightarrow \mathbb{R}^n$ has an $n \times n$ matrix w/ 1's on diagonal

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbb{R}^3 \rightarrow \mathbb{R}^3$$

Suppos $L: \mathbb{R}^6 \rightarrow \mathbb{R}^3$ is linear

$(L \circ I)(x) = L(x)$ so I is 6×6 matrix

$$\mathbb{R}^6 \xrightarrow{I} \mathbb{R}^6 \xrightarrow{L} \mathbb{R}^3$$

$(I \circ L)(x) = L(x)$ I is 3×3

$$\mathbb{R}^6 \xrightarrow{L} \mathbb{R}^3 \xrightarrow{I} \mathbb{R}^3$$

The set of linear functions $A \rightarrow B$ has a left identity for composition and a right identity for composition but they may not be the same

given $L: A \rightarrow B$ is there a K such that $(K \circ L)(x) = I(x)$?

The inverse of a linear function exists only if it is surjective and injective

Most linear functions are not invertible

given a matrix M for a linear function

$$L: A \rightarrow B$$

how do we compute its inverse?

how do we know it has an inverse?

$$\begin{array}{ccc} A & \xrightarrow{L} & B \\ \uparrow & & \downarrow \\ & & A \end{array} \quad \begin{array}{c} \xrightarrow{L^{-1}} \\ \xrightarrow{I} \end{array}$$

Gaussian Elimination

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

allowed moves

- 1) multiply any row by non zero const
- 2) add/subtract 1 row from another
- 3) swap positions of 2 rows

note! these moves change the linear function represented by the matrix. but it does not change the fact that the matrix is/isn't invertible